

Mechanistic equivalence and measurement design in models of bacterial nitrogen signaling

Yule Choi^{1*}

¹ Korea Science Academy of KAIST, Busan, Republic of Korea

* 25-117@ksa.hs.kr

Short title: Equivalence and measurement in nitrogen signaling

Abstract

A biochemical response can be precisely measured yet remain mechanistically ambiguous. We connect this ambiguity to concentration choice and observation design in bacterial nitrogen signaling. First, explicit GlnD and GlnE binding–catalysis transformations preserve complete stationary target-state distributions or downstream responses. Second, protein conservation bounds the differences between total observations of these free-state-equivalent mechanisms. A concurrent PII modification measurement tightens the cascade bound without requiring independent PII sites or a known downstream catalytic gain. In a source-anchored prospective example, the general bound is 0.370, whereas the observation-conditioned bound is 0.041, sufficient for a prescribed 0.05 tolerance after reducing GlnE fivefold. Third, an end-to-end binding benchmark propagates finite ligand depletion, shared baselines, concentration errors and estimated halfpoints into model discrimination. Independent concentration controls correct a tested systematic bias; an unmodeled binding-state mixture can still cause false mechanistic attribution despite small fitting residuals. Biochemical constraint profiles also show that GlnD occupancy contrast need not survive improved titration agreement. These results turn specified mechanistic alternatives into testable interpretation limits and measurement choices. They do not identify an unrestricted native mechanism or validate the complete physiological cascade. Published paired-paralog data motivate a separate prospective application whose calibration transfer remains unvalidated.

Author summary

Different molecular mechanisms can give the same protein response, even when every modification state is measured. We use bacterial nitrogen signaling to determine what experiments can still establish in that situation. Our first result constructs mechanisms that target measurements cannot distinguish. Our second result uses a measured PII modification count and protein conservation to determine when their total GS responses must remain close. In a worked example, the added observation supports a practical enzyme-concentration choice that a general bound cannot certify. Our third result tests binding measurements from individual concentration points through curve fitting to a final decision. The analysis shows why concentration controls can matter more than adding replicates, and why a well-fitted binding curve can still support the wrong molecular explanation. The calculations use reported biochemical scales, with assay-transfer assumptions kept explicit. They are predictions for controlled experiments, rather than new biological observations. The central contribution is a way to decide which conclusions survive mechanistic ambiguity and which additional measurements are needed.

Introduction

Ultrasensitivity describes a sharp input–output response, but the shape of that response does not uniquely identify its molecular source. Enzyme saturation, interactions among modification sites, reciprocal regulation, and downstream recognition can produce similar titrations [1, 2]. A useful mechanistic analysis must therefore answer two questions: which response properties follow from the reaction architecture, and which observations distinguish competing explanations of the same measured response?

The *Escherichia coli* nitrogen-signaling cascade provides a concrete setting for both questions. Glutamine regulates the bifunctional enzyme GlnD, which adds and removes UMP groups on trimeric PII. PII modification states regulate GlnE, which adds and removes adenylyl groups on the twelve-subunit glutamine synthetase (GS) complex [3–5]. Both layers combine multisite targets with opposing activities of a shared enzyme. Fig 1 summarizes the regulatory directions and state classes.

Fig 1. The GlnD–PII–GlnE–GS nitrogen-signaling cascade. Low free glutamine favors the UTase activity of GlnD, increasing PII uridylylation; uridylylated PII favors the adenylyl-removing (AR) activity of GlnE, driving GS toward lower adenylylation. High free glutamine favors the UR activity of GlnD, increasing unmodified PII; unmodified PII favors GlnE ATase, driving GS toward higher adenylylation. PII trimers carry 0–3 UMP groups, and GS dodecamers carry 0–12 adenylyl groups. The schematic summarizes regulatory direction and state classes rather than a kinetic model.

Published measurements report a glutamine response coefficient near two for the GlnD–PII cycle and approximately 4.5–6.5 for the full cascade, depending on PII concentration [6, 7]. Proposed explanations include allosteric regulation and bifunctionality [8, 9], zero-order and multistep effects [7], and selective GlnE activation by fully unmodified PII [6]. Bifunctionality can suppress sensitivity in some architectures but permit it through sequestration or additional enzyme–substrate complexes in others [10–13]. Reconstituted-cycle studies also show that concentration robustness depends on strong glutamine inhibition of UTase and on the enzyme construct [14]. These alternatives make the observation rule and assay conditions central to any quantitative attribution.

We organize the analysis around three results: regulatory transformations that preserve target distributions, an observation-conditioned bound that changes a concentration decision, and a binding-assay benchmark that tests discrimination from raw measurements. Stationary elimination, response decomposition and scaling non-identifiability are established foundations [11, 15–24]. The contribution is their biochemical construction and an explicit account of which measurements make the resulting bounds and decisions informative.

Results

Complete target distributions can leave enzyme mechanisms unresolved

The mathematical foundation is a stationary modification ladder. For a distributive target with n modification sites, a shared bifunctional enzyme, and regulatory exchange restricted to free enzyme, elimination of catalytic complexes gives

$$p_i = \frac{c_i u^i}{\sum_{j=0}^n c_j u^j}, \quad \theta = \frac{1}{n} \sum_i i p_i.$$

Here u is the ratio of opposing catalytic drives and $c_i > 0$ describes the effective ladder. This reduction uses established stationary elimination and rational parameterization [16–19]. Differentiation gives $d\theta/d\log u = \text{Var}(i)/n$. Consequently, local sensitivity separates into a ladder factor and switch elasticity. Independent identical sites give $c_i = \binom{n}{i}$ and $\theta = u/(1+u)$ for every n . Neither site count nor enzyme saturation alone steepens that free-target response. S1 Appendix derives the result and its restrictions; S1–S2 Figs retain the concentration and estimator comparisons.

This distinction matters for the twelve digitized GlnD–PII measurements [6]. The bounded fit has exponent 2.018 but a physical-occupancy midpoint local coefficient of 1.725. Those are different summaries of the same curve, not interchangeable measurements of cooperativity. Independent and interacting ladders fitted with the same four-parameter switch give RMSEs of 0.0352–0.0392 UMP groups per trimer. Exact inverse constructions can also preserve the mean while changing state fractions or introducing a refractory population. Downstream gain calibration can preserve some of those ambiguities. The titration, plateau, shared-parameter and downstream comparisons are retained in S3–S4 Figs, S2 Appendix and the repository extended-analysis archive; they compare stated models, not experimental probabilities of their native correctness.

A stronger ambiguity survives measurement of every target state. GlnD domain experiments motivate separate catalytic and regulatory functions [25–27], without establishing a unique assignment of model states to ACT domains. Let three free enzyme states have weights $w = (1, G/K_1, G^2/(K_1 K_2))$ and UT/UR catalytic specificities a_j, b_j . The stationary target drive depends on A/B , where $A = \sum_j a_j w_j$ and $B = \sum_j b_j w_j$. The transformation

$$K_1 \mapsto \lambda K_1, \quad K_2 \mapsto K_2/\lambda, \quad (a_1, b_1) \mapsto \lambda(a_1, b_1)$$

preserves both numerators: the singly liganded weight and its specificity compensate exactly. It therefore

preserves every free PII modification-state fraction at every free glutamine concentration, not merely a fitted mean (Fig 2).

A reference independent-ligand model fitted to the twelve coordinates has RMSE 0.0366 UMP groups per trimer and binding scale $K = 0.0520$ mM. Its transported UR specificity-fold scenario is stated in A5–A6 and D4; these coefficients are not independently measured microscopic rates. Transformations with $\lambda = 1, 2, 4$ preserve monotonically decreasing UT and increasing UR capacities but predict singly liganded fractions $1/2, 1/3, 1/5$ at $G = K$. Under rapid binding and unsaturated conversion, their relative relaxation rates are $1, 4/3, 8/5$. Average ligand occupancy at that input remains one for all three: measuring an average at a symmetry point can miss the very difference revealed by a resolved enzyme state.

Fig 2. GlnD mechanisms with identical PII states predict different enzyme occupancies and kinetics. (A) Three finite binding-and-catalysis constructions preserve the same fitted mean and every binomial PII state fraction. Points are the digitized coordinates. (B) Singly liganded enzyme fractions differ, reaching 0.5, 0.333, and 0.2 at the common binding scale. (C) PII relaxation-rate ratios in the rapid-binding, unsaturated conversion reduction; both capacities share the same multiplier. (D) An illustrative glutamine decrease from 0.5 to 0.05 mM gives different PII time courses with the same endpoint. Time is in reference-model units, not experimentally estimated minutes. The baseline is fitted once; the other models are obtained by the exact transformation in A6.

Good titration agreement is not sufficient biochemical validation. Transporting both the unliganded UT/UR specificity ratio and UR activation fold from the 1998 assays gives RMSE 0.540 in a constant-capacity model. Allowing different total catalytic capacities across regulatory states reduces this to 0.0409, but pushes a sequential binding scale to the fit boundary, about 148 mM. A fixed- K_1 profile shows the consequence for design: from $K_1 = 0.3$ to 10 mM, RMSE improves from 0.0493 to 0.0411 while the maximum singly liganded occupancy contrast within the monotone transformed family falls from 0.242 to 0.0108 (Fig 4D). These are cross-assay stress hypotheses, not confidence intervals. The 1998 UT and UR assays use 25 and 50 mM MgCl_2 , respectively, compared with 10 mM in the 2011 cascade assay [6, 26]. D4 documents the transfer conditions. Equivalence persists in this profile, but the original occupancy gap is not a robust biochemical prediction.

The published six-state GlnE topology supplies a downstream counterpart [28]. With $g = G/K_G$, $p = P/K_P$ and $u = U/K_U$, its binding polynomial is $Z_E = 1 + g + p + u + pg/\alpha_1 + pu/\alpha_2$. The stationary ATase/AR drive is

$$v = \frac{k_P}{k_R} \frac{p + \beta g + \beta' pg/\alpha_1}{u}.$$

The common binding denominator cancels. Thus α_2 is silent in this ratio, while common scaling of α_1 and β' trades binding synergy against catalysis without changing it. GlnE variants coupled to the GlnD family can preserve the GS response without condition-specific recalibration. They predict different PII-binding

halfpoints and reduced relaxation rates (S5 Fig). The direct glutamine term βg is retained: the cascade response is not attributable to PII alone.

These are constructive applications of binding-polynomial and scaling ideas [15, 23, 24], not a claim that stationary invariants or non-identifiability are new. Their specific contribution is an explicit biochemical family that defeats increasingly resolved target measurements. A state-weight change $w_j \mapsto s_j w_j$, compensated by $k_j \mapsto k_j/s_j$, preserves the corresponding catalytic numerator under the stipulated topology. A15 of S1 Appendix compares each result with prior theory. The construction neither exhausts equivalent mechanisms nor proves uniqueness of a native interpretation.

Conservation converts free-state equivalence into bounds on total observations

Experiments often count enzyme-bound target, whereas the invariant concerns free target. Finite mass-action binding need not destroy the stationary identity: complex balances recover the same free ratio when regulatory exchange is restricted to free enzyme. Total fractions can nevertheless differ. With one target per enzyme complex, two mechanisms sharing the free fraction and conserved totals satisfy

$$|\Delta\theta_{\text{total}}| \leq \min(1, E_T/T_T).$$

This is a conservation bound, not a low-occupancy approximation or a Hill-slope bound. Finite binding changes transients; regulatory exchange on catalytic complexes can change the stationary identity itself. S6 Fig separates these effects.

We then retain four PII states, thirteen GS states, three GlnD regulatory states and a refinement of the GlnE classes in a 120-species mass-action cascade. Catalytic complexes conserve PII, GS and both enzymes. Partly modified PII recognition is specified and varied as a model assumption; glutamine and nucleotide pools are buffered. Let P_T, S_T, D_T, E_T denote total PII trimers, GS dodecamers, GlnD and GlnE. Define

$$\epsilon_P = (D_T + 2E_T)/P_T < 1, \quad \epsilon_S = \min(1, E_T/S_T).$$

One GlnD can bind at most one PII and one GlnE at most two. The total-variation distance between total PII distributions in the equivalent pair is therefore at most ϵ_P , while their free PII distributions coincide.

Downstream, the specified GlnE drive has the form $r(T) = a + b/T$ for free PII pool size T , with $a, b \geq 0$. The direct glutamine route supplies b . Combining the permitted free-pool range with a ladder sensitivity

bound gives

$$|\Delta\theta_{GS,\text{total}}| \leq \min \left\{ 1, \tanh \left[\frac{\nu_*}{4} \log \frac{1}{1 - \epsilon_P} \right] + \epsilon_S \right\}.$$

Independent GS sites permit $\nu_* = 1$; $\nu_* = 12$ covers any positive twelve-site ladder shared by the pair.

Without the direct route, $b = 0$, free GS fractions coincide and only ϵ_S remains. These conclusions require the stated recognition and reaction structure (A10–A11).

The general bound can be very conservative: a small difference in one fitted realization does not establish a uniform interpretation guarantee. A concurrent observation can tighten it. Suppose the total PII mean UMP count has a justified upper bound M . The free zero-UMP fraction satisfies

$$\pi_0 \geq p_{\min} = \max\{0, 1 - M/(1 - \epsilon_P)\}.$$

This follows from nonnegative modification in the bound pool and requires no independent PII-site assumption. At fixed glutamine the source GlnE drive can be written $r(T) = k(a + b/T)$. A common positive factor k , including the AR gain, cancels from the drive contrast. With $g = G/K_G$, define

$$\ell = \frac{b}{a} \leq \frac{\beta g K_P}{p_{\min}(1 + \beta' g / \alpha_1)}, \quad d = \log \frac{1 + \ell/[P_T(1 - \epsilon_P)]}{1 + \ell/P_T}.$$

Then $|\Delta\theta_{GS,\text{total}}| \leq \min\{1, \epsilon_S + \tanh(\nu_* d/4)\}$. If $p_{\min} = 0$, the generic bound is recovered as $\ell \rightarrow \infty$. A11 proves the result and specifies its reaction assumptions.

Consider a prospective assay at $P_T = 5$, $S_T = 0.5$, $D_T = 0.5$, $E_T = 0.02 \mu\text{M}$ and $G = 2 \text{ mM}$, with a concurrently verified count bound $M \leq 0.67$. The source GlnE fit gives $K_G = 15.6 \text{ mM}$, $K_P = 0.18 \mu\text{M}$, $\alpha_1 = 0.17$, $\beta = 175/7.5$ and $\beta' = 1315/7.5$ [28]. For every positive twelve-site GS ladder shared by the pair, the generic bound is 0.370, while the count-conditioned bound is 0.0412. The latter certifies a prescribed 0.05 tolerance (Fig 3A). At $E_T = 0.10 \mu\text{M}$, the GS sequestration allowance alone is 0.20. Thus reducing GlnE fivefold, without also reducing GlnD, is sufficient in this declared example. This is a pairwise difference guarantee within the stated model class, not an error bound relative to biological truth or a globally optimal assay.

The numerical gate is motivated by the historical digitized mean 0.65 at 2 mM, but is not a confidence limit inferred from it. It must be measured in the proposed assay. Likewise, the source coefficients are a fitted example rather than independent microscopic measurements. At the proposed totals, an independently justified $\ell \leq 0.142 \mu\text{M}$ would suffice; the nominal value is 0.0162. Using an apparent-affinity sensitivity scale $K_P = 1 \mu\text{M}$ gives a bound of 0.0464. The source supplementary legend instead prints $K_P = 0.18 \text{ mM}$,

inconsistent with the main text and its stated direct-scale comparison; literal use gives 0.300 and loses the guarantee. Figure 3B shows this dependence explicitly. If the observation or coefficient bounds cannot be justified, the general bound must be used.

Five thousand arbitrary positive ladders and 5,000 free/bound PII mixtures verify the component inequalities. An additional 120 paired full-reaction realizations, using the source GlnE rate ratios and varied unknown gains, lie below the bound (Fig 3C); maximum stationary vector-field residual is 1.23×10^{-15} . These are finite-model checks, not validation of native kinetics or equilibration times. The earlier 8,200-condition sweep and topology checks remain in A10–A11 and S1 Data.

Fig 3. An observed PII count changes which concentration choices can be certified. (A) General and count-conditioned upper bounds on pairwise total GS fraction difference versus GlnE total. $P_T = 5$, $S_T = 0.5$, $D_T = 0.5 \mu\text{M}$, $G = 2 \text{ mM}$, $M \leq 0.67$, and the source-fit GlnE coefficients are fixed. Dashed horizontal line: prescribed 0.05 tolerance. (B) Bound at $E_T = 0.02 \mu\text{M}$ versus the count upper limit and K_P ; the contour marks 0.05. Markers identify the main-text source unit, an apparent-scale sensitivity scenario and the conflicting literal supplementary unit. (C) Actual differences and their general/count-conditioned bounds in 120 paired 120-species realizations; GlnD $\lambda = 1$ versus 4, GlnE $\sigma = 1$, three glutamine doses, two K_P values, five gains and four GlnE totals. Each bound uses the larger total PII count of the pair. These selected checks need not saturate a uniform bound. (D) Retained topology control: allowing EPU to catalyze adenylylation makes α_2 observable. Curves compare $\alpha_2 = 2.17$ and 8.85, maximizing over 41 specified doses at each added productive coefficient. All panels are conditional model calculations.

The boundary depends on topology and input control. Allowing the previously silent PII–PII-UMP GlnE class to catalyze adenylylation makes its interaction parameter observable; the selected extension reaches a GS fraction separation of 0.154 (Fig 3D). Yet the compensated family retains a drive $a + b/T + cT$ with log-elasticity magnitude at most one, so the general fraction bound survives. Fixing total instead of free glutamine is another intervention: different enzyme occupancies can produce different free inputs, with $0 \leq G_T - G \leq 2D_T + E_T$ in this network. These examples identify assumptions whose alteration changes the inference; they do not establish an additional native catalytic route.

Complementary measurements address distinct ambiguities

The next measurement should target what remains unresolved. State-resolved PII distinguishes the stipulated ladder and population alternatives. The adjacent-state contrast

$$\frac{p_{i-1}p_{i+1}}{p_i^2} = \frac{c_{i-1}c_{i+1}}{c_i^2}$$

is the established distributive-modification invariant [29]; here its input dependence tests a fixed homogeneous ladder. Rare-state ratios are error-sensitive. For four mean-matched candidates, the minimum

pairwise state-distribution distance is largest at 0.747 mM on the disclosed grid. A 10 mM measurement separates the independent and refractory scenarios' predicted fully modified fractions (S4 Fig). These are candidate-dependent designs, not experimentally established optima.

Such state measurements cannot separate the GlnD family. In the illustrative independent-binding family, a target-free measurement resolving singly liganded GlnD gives a minimum gap of 0.244 under a factor-of-two binding-scale variation. The kinetic-constraint profile above shows why this gap cannot be transported to every acceptable fit. Moreover, the assay must distinguish zero, one and two bound glutamines, rather than their mean. Intact native MS could be evaluated in a pilot, but successful GlnD state resolution and precision are not established here. The central observation benchmark therefore uses GlnE binding curves. Table 1 distinguishes the evidence and observations needed for these decisions.

With PII-UMP absent, normalized GlnE PII-binding halfpoints obey

$$\frac{H(G)}{H(0)} = \frac{1 + G/K_G}{1 + G/(\alpha_1 K_G)}.$$

The unknown PII affinity cancels. At appreciable enzyme concentration, a total-PII halfpoint equals $H(G) + E_T/2$ in the single-binding-class model. Two nonzero glutamine conditions plus a shared zero-glutamine curve identify K_G and α_1 ideally, except at $\alpha_1 = 1$ (D5). A solution assay proportional to bound GlnE fraction would require free and saturating-ligand controls, concentration standards and checks for glutamine-dependent signal changes.

We test the complete observation process at the source enzyme scale $E_T = 0.04 \mu\text{M}$: eleven total-PII doses, three glutamine concentrations and two technical readings per dose (Fig 4A). Each trial comprises eight independent complete three-condition titration sets. Curves are fitted jointly with finite ligand depletion, shared background and gain; a one-sided test uses the sample variance of each experiment's mean log-halfpoint ratio. This retains the covariance from the shared baseline. The reference is $\alpha_1 = 0.17$, with $K_G \in [7.8, 15.6]$ mM spanning the source's direct-scale comparison and joint-fit value. It is a declared nuisance range, not a measured confidence region. Noise, stock-calibration and enzyme-total errors are specified sensitivity scenarios (D5).

At the least-favorable tested null boundary, 100 Monte Carlo trials give rejection rates 0.06 for ideal calibration and 0.03 with shared stock and enzyme-total errors. Adding a systematic -0.10 log error to the nonzero-glutamine PII doses increases rejection to 0.96; independent concentration standards reduce it to 0/100 trials (95% Monte Carlo upper limit 0.037) in the tested correction scenario (Fig 4C). Under that correction, rejection increases across the tested small changes in α_1 (Fig 4B). Replicates denote complete

titration sets, not individual wells, and the plotted intervals report Monte Carlo uncertainty.

A misspecified two-affinity-class generator is a stronger limitation. Its condition-dependent class weights can produce rejection of the single-class reference while curve residuals remain similar to the correctly specified scenarios. Such rejection is evidence against the combined binding and calibration hypothesis; it cannot by itself identify a change in α_1 . This motivates concentration controls and tests of binding-state heterogeneity before interpreting a fitted halfpoint mechanistically. The earlier known-variance sample-size calculations remain illustrative special cases in D5, rather than the principal performance claim.

Fig 4. Binding measurements must survive observation error and biochemical constraints. (A) One simulated three-condition GlnE binding experiment, with two technical readings at each of eleven total-PII doses; lines show the finite-depletion fit. (B) Rejection rate versus the true regulatory parameter in the independently corrected dose-bias scenario; colors denote three K_G values. Each point summarizes 100 trials of eight independent complete three-condition titration sets. Intervals are 95% Wilson Monte Carlo intervals. (C) Rejection of the $\alpha_1 = 0.17$ reference at $K_G = 15.6$ mM across specified calibration and misspecification scenarios. Points and vertical intervals have the same meaning as B; the dashed line marks nominal size 0.05. (D) Strict cross-assay GlnD specificity transfer with nonconstant catalytic capacities: fixed- K_1 profile RMSE and maximum singly liganded midpoint contrast within each monotone equivalence family. Improved titration agreement can accompany diminished occupancy contrast. All panels are simulations or model fits; none is an experimentally validated assay budget.

Table 1. Evidence, observables and conditions for the central decisions.

Evidence class	Quantity supplied	What the inference additionally requires
Published observations	Twelve mean-UMP coordinates; source assay compositions and fitted GlnE coefficients	Cross-assay transfer is not automatic; rounded coordinates supply no biological confidence interval.
Exact conditional result	Free-state equivalence; count-conditioned total-GS bound	Stated reaction topology, common free input, conserved protein units and an upper coefficient bound.
Prospective observation	Total PII count M in the proposed cascade assay	Concurrent calibration; $M \leq 0.67$ is a gate to check, not an observed confidence limit.
Synthetic performance	Raw GlnE binding curves through halfpoint estimation and testing	Proportional binding readout, concentration controls and a tested binding-state model.
Prospective extension	Paired-paralog calibration and sequestration intervals	Matched calibration transfer and independent direct binding validation; not supplied by the cellular time series.

Prospective extension: a second paralog as a sequestration reporter

Independent in vivo data measure both modified and unmodified GlnB and GlnK peptides [5]. Their trajectories separate after nitrogen upshift (Fig 5A), but peptide measurements do not resolve intact trimer states. Purified AmtB binds fully deuridylylated GlnK, and ATP and 2-oxoglutarate regulate this interaction [30]. This supplies a kinetic-model-independent constraint before calibration: if AmtB-bound GlnK subunits are unmodified, their fraction s_K obeys $s_K \leq 1 - q_K$, where q_K is the total modified-site

fraction. For a homotrimer population, s_K is also the bound-trimer fraction and $s_K \leq p_0 \leq 1 - q_K$. The protein-level ceiling requires neither stationarity nor independent sites, but does require calibrated peptide measurements and the stated binding selectivity. At 15 min before upshift, the paired-peptide point fraction is 0.968, giving a point ceiling of 0.032. The working simultaneous intervals for the component means relax it to 0.162; using a separately measured total instead gives 0.668 (Fig 5D). Thus the available data can constrain binding under the observation model, while the choice and precision of normalization materially affect that constraint. These are upper limits, not occupancy estimates. A second target and a controlled calibration can sharpen the generic limit as follows.

Assume stationary independent-site free ladders, an unsequestered GlnB reporter, constant relative catalytic specificity ρ , and binding restricted to unmodified GlnK. With free GlnB modified fraction q_B , total GlnK modified fraction q_K , and bound GlnK trimer fraction s ,

$$F_\rho(q_B) = \frac{\rho q_B}{1 - q_B + \rho q_B}, \quad s = 1 - \frac{q_K}{F_\rho(q_B)}.$$

The common GlnD drive cancels. A control without AmtB binding calibrates ρ from the ratio of modification odds, provided relative specificities are unchanged. A defined reconstitution permits tighter input control than a cellular deletion. Without calibration, the means do not identify s : unrestricted ρ leaves only the generic limit $s \leq 1 - q_K$.

Monotonicity propagates control and assay intervals into deterministic sequestration intervals. In the disclosed synthetic example these give $s \in [0.441, 0.561]$; allowing an additional absolute free-relation error of 0.10 widens the interval to $[0.347, 0.610]$ (Fig 5C). They are not experimental confidence intervals. A13 retains the inputs, derivation and independent generator checks.

Fig 5. Paired paralogs motivate a calibrated sequestration assay. (A) Published wild-type paired-peptide fractions around nitrogen upshift; error bars are conservative first-order SE bounds with unknown covariance. Connecting lines do not establish steady state. (B) Conditional free-pool relation at three relative-specificity values. (C) Synthetic bound-fraction intervals with the control and assay ranges in the text, allowing an additional absolute error in the free-pool relation. Shading is an admissible interval, not a confidence band. (D) Upper bounds $s_K \leq 1 - q_K$ on the bound GlnK protein fraction. This is also a bound-trimer fraction for homotrimers; it does not identify mixed-particle counts. The dashed curve uses paired-peptide point fractions; solid curves use working Bonferroni-adjusted t intervals for three component means at each time point, under paired-peptide or separate-total normalization. They do not have established simultaneous coverage across the time course and do not include calibration bias. These ceilings require no stationarity; the calibrated reconstruction in B–C does. None of the curves is a measured bound fraction.

The odds formula extends to any positive free ladders sharing one scalar drive: monotonicity defines an increasing relation $C = m_K \circ m_B^{-1}$, which a no-binding titration can calibrate. Bounds on reporter partition, free-pool relation error and imperfect binding selectivity can then be propagated by a sharp mixture

calculation (A14). For example, free GlnK fractions $[0.70, 0.80]$ and total fractions $[0.35, 0.39]$ imply $s \in [0.443, 0.563]$ if the bound pool is unmodified; permitting its modified fraction to reach 0.15 widens this to $[0.443, 0.692]$. The full inequalities and degeneracies remain in A14.

The biological validation requirement is separate. Unmodifiable GlnK Y51A changes membrane association after ammonium shock despite unchanged modification [31], and GlnB/GlnK heterotrimers occur in vivo [32]. Thus modification alone need not identify partition or particle counts. The available cellular time series supplies neither a matched calibration nor direct occupancy validation. Native MS and surface plasmon resonance provide independent AmtB–GlnK binding endpoints [33], but measurements under matched conditions are needed to test calibration transfer. Figure 5 is a prospective application, not a validated biological reconstruction.

Discussion

The main result is an operational distinction between what equivalent mechanisms permit us to interpret and what they permit us to identify. Complete target-state resolution can leave regulatory compensation unresolved. Protein conservation and a concurrent PII count can nevertheless certify that the total GS predictions remain close, even when a general capacity bound is uninformative. A binding assay can address the remaining compensation, provided its own observation process is controlled.

Stationary elimination, binding-polynomial derivatives, adjacent-state invariants and scaling non-identifiability are established [15–18, 20, 21, 23, 24, 29]. The methodological advance here is their composition into a biochemical decision case: an observed modification count supplies a coefficient bound that supports an enzyme-concentration choice across unknown downstream gains and positive GS ladders. This differs from selecting the small error of one fitted realization. A15 compares the results with their theoretical antecedents.

The binding benchmark demonstrates both a useful correction and its limit. Independent concentration standards correct the tested systematic dose bias; replication alone cannot remove that bias. A hidden affinity mixture can still produce false attribution within the single-class interpretation despite a good fit. Likewise, tighter agreement with transported GlnD catalytic ratios can greatly reduce the occupancy difference targeted by a proposed assay. These results make assay controls and candidate constraints part of the inference rather than annotations to a sample-size calculation.

The biological scope remains defined by reaction topology and observation. Productive regulatory complexes and conserved rather than free glutamine can alter the equivalence; hidden conformations can

defeat complementary binding measurements. The finite network does not validate the complete physiological cascade, and lowering enzyme concentration requires renewed checks of equilibration. ATP/ADP, UTP, metabolism and expression feedback are outside the core stationary analysis. Count states omit spatial modification arrangements that can matter in other multisite systems [34]. Glutamine:2-oxoglutarate regulation in another species motivates controlling both inputs, but is not an E. coli parameter constraint [35].

The worked certificate therefore specifies measurements that remain to be made: a concurrent PII count bound, a justified GlnE coefficient envelope and stationarity under the chosen totals. The binding benchmark requires readout and concentration calibration and tests for heterogeneous binding. The paired-paralog extension additionally requires a matched no-binding calibration and an independent bound-fraction endpoint. Historical digitization and cellular time series do not supply those missing validations. The present evidence supports a conditional methodology for interpretation and experiment design, rather than selection of a unique native mechanism.

Materials and methods

Reaction models and verification

Core models use positive mass-action rates, distributive catalytic release and explicitly stated regulatory exchange. Complex balances and zero flux across ladder cuts give the stationary distributions; derivatives and conservation yield the response and observation bounds. S1 Appendix supplies complete reaction schemes, proofs, parameter origins and boundary constructions. S2 Appendix supplies data provenance, fitting sensitivity and prospective assay precision; earlier exploratory analyses are preserved in repository documentation. Shared rate-rescaling constructions are not fitted independently to the same data and are not assigned model-selection probabilities.

The closed cascade enumerates 120 species and 300 directed reactions; an additional productive route gives a 156-species model. Stationary solutions reduce to conserved free-pool balances and are checked in the independently enumerated mass-action vector field. Conserved-input calculations add a scalar glutamine balance. Randomized comparisons, alternative recognition weights and independent BDF integrations test specified realizations. Dimensions, tolerances, seeds and parameter ranges are recorded in A10–A12 and S1 Data.

A source copy with generated results removed completed all sixteen regeneration and verification stages in an isolated virtual environment, including the baseline, all mechanistic extensions and original-workbook

extraction. All 22 baseline checks, matched-model verification and three extension test groups passed. The integrated entry point additionally checks 2,000 independent paired-reporter generators and 10,000 admissible interval configurations. The generalized calibration extension is checked against 3,000 independent mixture linear programs, 500 reporter-interval linear programs and 1,000 positive ladder pairs. Machine-readable reports distinguish clean regeneration from subsequent checks. They validate computational statements under the model assumptions, not experimental uncertainty or physiological correctness.

Published data and comparison rules

The first-layer analysis uses twelve author-supplied digitized coordinate pairs, retained with provenance. Bounded Hill and specified ladder fits use identical observations. Corrected AIC includes the residual variance parameter under a working independent Gaussian model; profile calculations follow established practice [36]. Coordinate perturbations use prespecified input/output scales, not estimated biological errors. Two thousand refits assess coordinate sensitivity and 300 shared perturbations propagate state predictions. Historical downstream summaries receive shared-parameter comparisons without inventing unreported titration coordinates or likelihoods. Numerical and source details are in S2 Appendix and the repository extended-analysis archive.

The independent Gosztolai et al. workbook and model are retained unchanged from doi:10.6084/m9.figshare.4880003. Extraction records source cells, means, SE and sample counts. Paired-peptide fractions divide modified signal by modified plus unmodified signal; they differ from normalization by a separately measured total. First-order SE uses the conservative sum of absolute contributions because covariance is unavailable. It does not establish joint confidence coverage. Raw copy-per-cell values are preserved; the source model corresponds to $0.00161\ \mu\text{M}$ per copy per cell, and the workbook's inconsistent printed unit is documented.

Prospective observation design

State-distribution designs maximize a disclosed candidate-dependent minimum distance. The GlnE observation benchmark generates finite-depletion binding measurements at three glutamine concentrations and eleven PII doses, fits three free halfpoints with a common background and gain, and tests the average of two log ratios using the variance across independent full experiments. The shared zero-glutamine fit is retained in both ratios. Scenarios include 10% log-scale stock error, 20% enzyme-total error, a condition-dependent log-dose bias of -0.10 , independent correction standards with log SD 0.02 , and a two-affinity-class generator fitted with one class. Signal noise and all seeds, doses, fitting bounds and

rejection rules are specified in D5 and S1 Data. The 100 trials per condition yield Monte Carlo intervals, not guaranteed frequentist performance for a native assay.

The count-conditioned bound is tested with arbitrary positive ladders, independently formed protein mixtures and full conserved reaction realizations. Paired-reporter intervals require their calibration and uncertainty sets to hold jointly. D4–D6 and A11–A14 separate reported values, source-fit examples, chosen scenarios and measurements required for application.

Data and code availability

No new experimental measurements were generated. Source observations, digitized coordinates, parameter records, prediction grids and verification outputs are supplied as S1 Data. The synchronized manuscript and analysis source are released as GitHub version 1.3.0, with the source manifest accompanying this submission. The public repository is github.com/choeyule-gif/nitrogen-signaling-ultrasensitivity. The earlier v1.1.0 archive, doi:10.5281/zenodo.22735182, does not contain the later paired-paralog or observation-design extensions. A DOI for the present revision will be added only after its archival record is verified. The independent source dataset is available at doi:10.6084/m9.figshare.4880003. S1 Data retains source licenses and attribution.

Acknowledgments

I would like to express my sincere gratitude to Dr Seonhwa Park for introducing me to the practice of research and for her guidance and encouragement.

Author contributions

Yule Choi: conceptualization, methodology, formal analysis, software, investigation, validation, visualization, writing – original draft, writing – review and editing.

Competing interests

The author declares no competing interests.

References

1. Goldbeter A, Koshland DE. An amplified sensitivity arising from covalent modification in biological systems. *Proc Natl Acad Sci USA*. 1981;78:6840–6844. doi:10.1073/pnas.78.11.6840.
2. Ferrell JE, Ha SH. Ultrasensitivity part III: cascades, bistable switches, and oscillators. *Trends Biochem Sci*. 2014;39(12):612–618. doi:10.1016/j.tibs.2014.10.002.
3. van Heeswijk WC, Westerhoff HV, Boogerd FC. Nitrogen Assimilation in *Escherichia coli*: Putting Molecular Data into a Systems Perspective. *Microbiol Mol Biol Rev*. 2013;77(4):628–695. doi:10.1128/mmmbr.00025-13.
4. Ninfa AJ, Atkinson MR. PII signal transduction proteins. *Trends Microbiol*. 2000;8:172–179. doi:10.1016/s0966-842x(00)01709-1.
5. Gosztolai A, Schumacher J, Behrends V, Bundy JG, Heydenreich F, Bennett MH, et al. GlnK Facilitates the Dynamic Regulation of Bacterial Nitrogen Assimilation. *Biophys J*. 2017;112:2219–2230. doi:10.1016/j.bpj.2017.04.012.
6. Jiang P, Ninfa AJ. A Source of Ultrasensitivity in the Glutamine Response of the Bicyclic Cascade System Controlling Glutamine Synthetase Adenylylation State and Activity in *Escherichia coli*. *Biochemistry*. 2011;50(50):10929–10940. doi:10.1021/bi201410x.
7. Ventura AC, Jiang P, Van Wassenhove L, Del Vecchio D, Merajver SD, Ninfa AJ. Signaling properties of a covalent modification cycle are altered by a downstream target. *Proc Natl Acad Sci U S A*. 2010;107(22):10032–10037. doi:10.1073/pnas.0913815107.
8. Mutalik VK, Shah P, Venkatesh KV. Allosteric Interactions and Bifunctionality Make the Response of Glutamine Synthetase Cascade System of *Escherichia coli* Robust and Ultrasensitive. *J Biol Chem*. 2003;278(29):26327–26332. doi:10.1074/jbc.m300129200.
9. Bruggeman FJ, Boogerd FC, Westerhoff HV. The multifarious short-term regulation of ammonium assimilation of *Escherichia coli*: dissection using an in silico replica. *FEBS J*. 2005;272(8):1965–1985. doi:10.1111/j.1742-4658.2005.04626.x.
10. Ortega F, Ehrenberg M, Acerenza L, Westerhoff HV, Mas F, Cascante M. Sensitivity Analysis of Metabolic Cascades Catalyzed by Bifunctional Enzymes. *Mol Biol Rep*. 2002;29(1-2):211–215. doi:10.1023/a:1020334014497.
11. Ortega F, Acerenza L, Westerhoff HV, Mas F, Cascante M. Product dependence and bifunctionality compromise the ultrasensitivity of signal transduction cascades. *Proc Natl Acad Sci USA*. 2002;99:1170–1175. doi:10.1073/pnas.022267399.
12. Straube R. Sensitivity and Robustness in Covalent Modification Cycles with a Bifunctional Converter Enzyme. *Biophys J*. 2013;105(8):1925–1933. doi:10.1016/j.bpj.2013.09.010.
13. Straube R. Reciprocal Regulation as a Source of Ultrasensitivity in Two-Component Systems with a Bifunctional Sensor Kinase. *PLoS Comput Biol*. 2014;10(5):e1003614. doi:10.1371/journal.pcbi.1003614.
14. Jiang P, Zhang Y, Atkinson MR, Ninfa AJ. The Robustness of the *Escherichia coli* Signal-Transducing UTase/UR-PII Covalent Modification Cycle to Variation in the PII Concentration Requires Very Strong Inhibition of the UTase Activity of UTase/UR by Glutamine. *Biochemistry*. 2012;51(45):9032–9044. doi:10.1021/bi3005736.
15. Adair GS, Bock AV, Field H Jr. The hemoglobin system: VI. The oxygen dissociation curve of hemoglobin. *J Biol Chem*. 1925;63(2):529–545. doi:10.1016/S0021-9258(18)85018-9.

16. Thomson M, Gunawardena J. The rational parameterisation theorem for multisite post-translational modification systems. *J Theor Biol.* 2009;261:626–636. doi:10.1016/j.jtbi.2009.09.003.
17. Feliu E, Wiuf C. Variable elimination in post-translational modification reaction networks with mass-action kinetics. *J Math Biol.* 2013;66(1-2):281–310. doi:10.1007/s00285-012-0510-4.
18. Feliu E, Wiuf C. Variable Elimination in Chemical Reaction Networks with Mass-Action Kinetics. *SIAM J Appl Math.* 2012;72:959–981. doi:10.1137/110847305.
19. Xu Y, Gunawardena J. Realistic enzymology for post-translational modification: Zero-order ultrasensitivity revisited. *J Theor Biol.* 2012;311:139–152. doi:10.1016/j.jtbi.2012.07.012.
20. Abeliovich H. An Empirical Extremum Principle for the Hill Coefficient in Ligand-Protein Interactions Showing Negative Cooperativity. *Biophys J.* 2005;89:76–79. doi:10.1529/biophysj.105.060194.
21. Briggs WE. Cooperativity and extrema of the Hill slope for symmetric protein-ligand binding polynomials. *J Theor Biol.* 1984;108(1):77–83. doi:10.1016/s0022-5193(84)80170-8.
22. Silk D, Kirk PDW, Barnes CP, Toni T, Stumpf MPH. Model Selection in Systems Biology Depends on Experimental Design. *PLoS Comput Biol.* 2014;10(6):e1003650. doi:10.1371/journal.pcbi.1003650.
23. Castro M, de Boer RJ. Testing structural identifiability by a simple scaling method. *PLoS Comput Biol.* 2020;16(11):e1008248. doi:10.1371/journal.pcbi.1008248.
24. Villaverde AF, Massonis G. On testing structural identifiability by a simple scaling method: Relying on scaling symmetries can be misleading. *PLoS Comput Biol.* 2021;17(10):e1009032. doi:10.1371/journal.pcbi.1009032.
25. Zhang Y, Pohlmann EL, Serate J, Conrad MC, Roberts GP. Mutagenesis and Functional Characterization of the Four Domains of GlnD, a Bifunctional Nitrogen Sensor Protein. *J Bacteriol.* 2010;192(11):2711–2721. doi:10.1128/jb.01674-09.
26. Jiang P, Peliska JA, Ninfa AJ. Enzymological Characterization of the Signal-Transducing Uridylyltransferase/Uridylyl-Removing Enzyme (EC 2.7.7.59) of *Escherichia coli* and Its Interaction with the PII Protein. *Biochemistry.* 1998;37(37):12782–12794. doi:10.1021/bi980667m.
27. Grant GA. The ACT Domain: A Small Molecule Binding Domain and Its Role as a Common Regulatory Element. *J Biol Chem.* 2006;281:33825–33829. doi:10.1074/jbc.r600024200.
28. Jiang P, Mayo AE, Ninfa AJ. *Escherichia coli* glutamine synthetase adenylyltransferase (ATase, EC 2.7.7.49): kinetic characterization of regulation by PII, PII-UMP, glutamine, and alpha-ketoglutarate. *Biochemistry.* 2007;46:4133–4146. doi:10.1021/bi0620510. Supporting information: doi:10.1021/bi0620510.s001.
29. Gunawardena J. Distributivity and processivity in multisite phosphorylation can be distinguished through steady-state invariants. *Biophys J.* 2007;93(11):3828–3834. doi:10.1529/biophysj.107.110866.
30. Durand A, Merrick M. In vitro analysis of the *Escherichia coli* AmtB-GlnK complex reveals a stoichiometric interaction and sensitivity to ATP and 2-oxoglutarate. *J Biol Chem.* 2006;281(40):29558–29567. doi:10.1074/jbc.M602477200.
31. Radchenko MV, Thornton J, Merrick M. Association and dissociation of the GlnK–AmtB complex in response to cellular nitrogen status can occur in the absence of GlnK post-translational modification. *Front Microbiol.* 2014;5:731. doi:10.3389/fmicb.2014.00731.
32. van Heeswijk WC, Wen D, Clancy P, Jaggi R, Ollis DL, Westerhoff HV, et al. The *Escherichia coli* signal transducers PII (GlnB) and GlnK form heterotrimeric in vivo: fine tuning the nitrogen signal cascade. *Proc Natl Acad Sci USA.* 2000;97(8):3942–3947. doi:10.1073/pnas.97.8.3942.

33. Cong X, Liu Y, Liu W, Liang X, Laganowsky A. Allosteric modulation of protein-protein interactions by individual lipid binding events. *Nat Commun.* 2017;8:2203. doi:10.1038/s41467-017-02397-0.
34. Callenbach A, Dorešić D, Düster R, Nakonecnij V, Dudkin E, Geyer M, et al. Quantitative modelling of P-TEFb mediated CTD phosphorylation identifies local cooperativity. *PLoS Comput Biol.* 2026;22(7):e1014531. doi:10.1371/journal.pcbi.1014531.
35. Emori MT, Tomazini LF, Souza EM, Pedrosa FO, Chubatsu LS, Oliveira MAS. The deuridylylation activity of *Herbaspirillum seropedicae* GlnD protein is regulated by the glutamine:2-oxoglutarate ratio. *Biochim Biophys Acta Proteins Proteom.* 2018;1866(12):1216–1223. doi:10.1016/j.bbapap.2018.09.009.
36. Raue A, Kreutz C, Maiwald T, Bachmann J, Schilling M, Klingmüller U, et al. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics.* 2009;25(15):1923–1929. doi:10.1093/bioinformatics/btp358.

Supporting information

S1 Appendix. Reaction mechanisms, conservation proofs, calibrated reporter bounds and prior-art comparison (A1–A15).

S2 Appendix. Data provenance, fitting sensitivity, biochemical constraints and prospective assay precision (D1–D7).

S1 Data. Source measurements, parameter records, prediction grids, code and verification reports.

S1–S6 Figs. Concentration dependence, ladder constraints, titration comparisons, state-resolved alternatives, GlnE compensation and finite-reaction boundaries, supplied separately with captions.